

MANIKANDAN RAJ

RESEARCH & DEVELOPMENT ENGINEER — ROBOTICS & INDUSTRIAL AUTOMATION

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PROFESSIONAL SUMMARY

Research & Development Engineer with 5+ years of combined experience across manufacturing, industrial automation, and robotics R&D. A practical, execution-focused engineer with hands-on exposure to Smart Factory implementation, AMR integration, CNC/VMC systems, and ROS2-based robotics. Strong in project coordination, system integration, technical troubleshooting, and documentation, supported by direct shop-floor manufacturing experience and a proven ability to handle client-facing technical demonstrations and training.

CORE COMPETENCIES

Smart Factory Implementation • AMR Integration Support • Industrial Automation • Robotics R&D Support • Project Coordination • System Integration • Technical Troubleshooting • Technical Documentation • Client & Vendor Coordination • Robotics Training & Demonstrations

PROFESSIONAL EXPERIENCE

Research & Development Engineer

JAN 2025 — PRESENT

Ausweg Info Controls Private Limited

Coimbatore, India

- Support implementation and execution of Smart Factory automation projects across manufacturing lines.
- Assist with AMR integration, robotics support, and industrial automation workflow setup.
- Work on CNC/VMC machine connectivity and MTLINK-i data integration for production monitoring.
- Contribute to RFID integration concepts and basic PLC/HMI development tasks.
- Coordinate project activities and support client interaction, vendor coordination, and technical documentation.
- Conduct robotics workshops and technical demonstration sessions for internal and client audiences.

Robotics R&D Engineer

JUL 2023 — JAN 2025

Robomiracle Technologies Private Limited

India

- Supported robotics and drone-based research and development activities.
- Gained hands-on exposure to ROS2-based robotics and autonomous system concepts.
- Assisted with humanoid robotics development and basic computer-vision tasks using Python and OpenCV.
- Delivered robotics training and STEM workshop sessions, and supported technical demonstrations.

Production In-charge — CNC / VMC Operations

NOV 2019 — JUL 2023

Mira Alloy Steels Private Limited

India

- Progressed from machine operator to production in-charge with full shop-floor responsibility.
- Handled CNC lathe and VMC machine setup, operation, and production coordination.
- Managed daily production workflow and built practical exposure to manufacturing systems and quality requirements.

TECHNICAL SKILLS

ROBOTICS & AUTOMATION ROS2 (integration exposure), AMR concepts, industrial robotics, Smart Factory systems

INDUSTRIAL SYSTEMS CNC / VMC machines, MTLINK-i, RFID systems, Delta PLC (intermediate), Siemens PLC (basic), HMI & SCADA (basic)

PROGRAMMING Python scripting (basic), OpenCV (basic), AI-assisted development

PROFESSIONAL Project coordination, technical documentation, troubleshooting, client & vendor interaction

EDUCATION

B.Sc — Information Technology

Alagappa University, Tamil Nadu

LANGUAGES

Tamil • English • Malayalam